

PROG101
PRINCIPLES OF PROGRAMMING LOGIC AND DESIGN

**Individual
Assignment**

Assignment Title : Designing Logical Solutions for Sierra Leone Community Challenges
Examiner : Elijah Fullah
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Issues Date : Week 2
Submission Deadline : Week 4

Prepared by: Elijah Fullah

Signature:

Date:

Checked by: Sheik Umar Kanu

Signature:

Date:

Approved by AQA:

Stamp, Date & Signature:

1. OVERVIEW

The Government of Sierra Leone and many local communities face challenges in areas such as healthcare, education, agriculture, transportation, waste management, flooding, and access to public services. Before developing software, every successful system begins with proper problem analysis and logical design.

You have been hired as a Junior System Analyst to design a logical solution for one community problem affecting Sierra Leone. Your task is not to write a computer program, but to design how the system should work using computational thinking, algorithms, pseudocode, flowcharts, and modular design.

Your solution should also demonstrate how Open Source Software and Digital Public Goods (DPGs) can help improve access to technology while contributing to the United Nations Sustainable Development Goals (SDGs).

2. ASSIGNMENT OBJECTIVES

By completing this assignment, students should be able to:

- I. Apply computational thinking to analyze community problems.
- II. Design efficient algorithms using pseudocode.
- III. Develop professional flowcharts using diagrams.net (Draw.io), Mermaid, or Flowgorithm.
- IV. Apply decision structures, repetition structures, modules, and functions within algorithm design.
- V. Demonstrate logical problem-solving techniques.
- VI. Apply Git and GitHub for version control and project management.
- VII. Explain how Open Source Software and Digital Public Goods contribute to solving societal problems.
- VIII. Develop solutions that directly address challenges in Sierra Leone and support one or more Sustainable Development Goals (SDGs).

3. APPLICATION REQUIREMENT (MANDATORY)

Students are **NOT** required to write source code for this assignment.

Instead, students must design the complete logical solution using:

- Algorithms
- Pseudocode
- Flowcharts
- Modular Design
- Functions
- Decision Structures
- Repetition Structures

4. SYSTEM DESCRIPTION (CHOOSE ONE ONLY)

Students must choose **ONE (1)** real-world problem affecting Sierra Leone and design a logical solution.

5. FUNCTIONAL REQUIREMENTS

5.1. Problem Analysis

Students must clearly describe:

- The community problem.
- Who is affected.
- Why the problem exists.
- How the proposed solution addresses the problem.

5.2. Algorithm Design

Students must prepare:

- Step-by-step algorithm
- Pseudocode
- Flowchart

5.3. Logical Design

The solution should demonstrate:

- Input
- Processing
- Output
- Decision Structures
- Repetition Structures
- Modular Design
- Functions

5.4. Output

Students should demonstrate the expected outputs using sample scenarios.

6. PROGRAMMING LOGIC REQUIREMENTS

The logical design must demonstrate:

- Computational Thinking

- Algorithm Design
- Variables
- Constants
- Data Types
- Decision Structures
- Boolean Logic
- Repetition Structures
- Functions
- Modular Design
- Problem Decomposition

7. DIGITAL PUBLIC GOODS, OPEN SOURCE AND SDG REQUIREMENTS

Each project **must**:

7.1. Sustainable Development Goals

Identify at least **ONE (1)** Sustainable Development Goal supported by the project.

7.2. Open Source Software

Explain:

- Why the proposed solution should be Open Source.
- Benefits of Open Source for Sierra Leone.
- Potential community contributions.

7.3. Digital Public Goods (DPGs)

Explain:

- How the proposed system could become a Digital Public Good.
- How it promotes accessibility, openness, and public benefit.

7.4. Sierra Leone Community Impact

Clearly explain:

- The problem being solved.
- The expected impact on the local community.
- The beneficiaries.
- Long-term sustainability.

8. DOCUMENTATION REQUIREMENTS (COMPULSORY)

Students must submit their work in a neat and professionally presented format mounted on a board.

8.1. Hardcopy Submission

Students must submit:

1. Cover Page
2. Table of Contents
3. Two to Three Page Rationale including:
 - a) Problem Description
 - b) Background
 - c) Community Impact

- d) Proposed Solution
- e) Computational Thinking Process
- f) Algorithm Design
- g) SDG Alignment
- h) Open Source Justification
- i) Digital Public Goods Justification
- 4. Algorithm
- 5. Pseudocode
- 6. Flowcharts
- 7. Functional Modules
- 8. Expected Outputs
- 9. References

All hardcopy materials must be:

- Printed clearly
- Arranged neatly
- Neatly bound.

8.2. GitHub Evidence (Included in Report)

Students must include evidence of:

- GitHub Repository
- Repository URL
- Git Commit History
- README.md
- Repository Screenshot

9. GITHUB SUBMISSION (MANDATORY)

Students must:

1. Create a GitHub account
2. Create repository using the naming convention:
◆ **PROG101_Assignment1_StudentName_StudentID**
3. Upload:
 - Report
 - Flowcharts
 - Pseudocode
 - README.md

Students should demonstrate the use of Git through regular commits during project development.

10. SUBMISSION INSTRUCTIONS

10.1. Hardcopy Submission

Submit a printed copy containing:

- Printed Report
- Flowcharts
- Pseudocode
- Algorithm Design
- Screenshots
- GitHub Evidence

10.2. Softcopy Submission

- Printed Report
- Flowcharts
- Pseudocode
- Algorithm Design
- Screenshots
- README.md

11. ASSESSMENT CRITERIA (100 MARKS)

Criteria	Marks
Problem Context & Community Relevance	10
Computational Thinking & Problem Analysis	10
Algorithm Design	15
Pseudocode	15
Flowchart Design	10
Logical Design (Decision Structures, Repetition, Modules & Functions)	15
SDG, Open Source & Digital Public Goods Integration	10
GitHub Repository & Version Control	5
Report Quality & Professional Presentation	10
Total	100

12. LATE SUBMISSION & ACADEMIC INTEGRITY

- Late submission attracts a **10-mark penalty**
- Plagiarism or shared repositories will attract disciplinary action
- Students must adhere strictly to the university's academic policies
- Use the following format:
 - **Font Type:** Tahoma
 - **Font Size:** 10
 - **Line Spacing:** 1.15
 - **Alignment:** Justify